

Prescribing a power spectrum from Fourier-space variance

Let u be a statistically isotropic velocity field on $\mathbb{R}^d$ (or on $\mathbb{T}^d$), with Fourier transform $\widehat{u}(\xi)$. Define the one-dimensional energy spectrum by

$$E(k) := \frac{d}{dk} \int_{\{|\xi| \leq k\}} \mathbb{E}|\widehat{u}(\xi)|^2 d\xi = \int_{\{|\xi|=k\}} \mathbb{E}|\widehat{u}(\xi)|^2 dS(\xi), \quad k > 0.$$

This definition is the same in every dimension. The dimension dependence enters only through the size of the shell $\{|\xi| = k\}$.

Assume the Fourier coefficients are generated by

$$\widehat{u}(\xi) = \sigma(|\xi|) g_\xi, \quad \mathbb{E}|g_\xi|^2 = 1,$$

so that $\sigma(k)^2 = \mathbb{E}|\widehat{u}(\xi)|^2$ for $|\xi| = k$. Then, on $\mathbb{R}^d$,

$$E(k) = |\mathbb{S}^{d-1}| k^{d-1} \sigma(k)^2.$$

Hence, if one wants

$$E(k) \sim C k^{-p},$$

one should choose

$$\sigma(k)^2 \sim C |\mathbb{S}^{d-1}|^{-1} k^{-(p+d-1)}.$$

Equivalently, the standard deviation of each Fourier coefficient on the shell $|\xi| = k$ should scale like

$$\sigma(k) \sim k^{-(p+d-1)/2}.$$

On a discrete Fourier grid, the same scaling holds up to constants, since the number of modes in a shell of radius k is $\asymp k^{d-1}$.

Kolmogorov example. If $E(k) \propto k^{-5/3}$, then $p = 5/3$ and therefore

$$\sigma(k)^2 \propto k^{-(d+2/3)}, \quad \sigma(k) \propto k^{-(d/2+1/3)}.$$

In particular,

$$\sigma(k)^2 \propto k^{-8/3} \quad (d=2), \quad \sigma(k)^2 \propto k^{-11/3} \quad (d=3).$$

Relation with Hölder regularity. Heuristically (and exactly in the standard Gaussian scaling models), if the velocity field has Hölder exponent α , then

$$\mathbb{E}|\widehat{u}(\xi)|^2 \sim |\xi|^{-(2\alpha+d)}.$$

After integrating over the shell $|\xi| = k$, this gives

$$E(k) \sim k^{d-1} k^{-(2\alpha+d)} = k^{-(2\alpha+1)}.$$

Thus, if

$$E(k) \sim k^{-p},$$

then the corresponding Hölder exponent is

$$p = 2\alpha + 1, \quad \alpha = \frac{p-1}{2}.$$

In particular, the Kolmogorov law $E(k) \propto k^{-5/3}$ corresponds to $\alpha = 1/3$.

All proportionality constants depend on Fourier normalization, shell thickness, and any additional constraints such as reality or divergence-free projection, but the exponents above are unchanged.